Ceramics International 48 (2022) 9927–9938

Contents lists available at ScienceDirect

Ceramics International

journal homepage: www.elsevier.com/locate/ceramint

Experimental phase equilibria studies in the “$CuO_{0.5}$”-CaO-SiO_2 ternary system in equilibrium with metallic copper

Georgii Khartcyzov *, Maksym Shevchenko, Siyu Cheng, Peter C. Hayes, Evgueni Jak

PYROSEARCH, Pyrometallurgy Innovation Centre, School of Chemical Engineering, The University of Queensland, Brisbane, Queensland, 4072, Australia

ARTICLE INFO

Keywords:
Phase equilibria
$CuO_{0.5}$
CaO
SiO_2
Copper
Liquidus
Refractories
Microstructure

ABSTRACT

Phase equilibria studies were undertaken on the “$CuO_{0.5}$”-CaO-SiO_2 system in equilibrium with metallic copper to provide fundamental information on the reactions taking place between slags and refractory materials in metallurgical furnaces. The liquidus isotherms between 1160 and 1706 °C in the tridymite and cristobalite (SiO_2), cuprite (Cu_2O), pseudowollastonite ($CaSiO_3$), rankinite ($Ca_3Si_2O_7$) and dicalcium silicate (Ca_2SiO_4) primary phase fields, and the 2-liquid miscibility gaps in the silica-rich corner and low-silica area of the system, were experimentally determined using the equilibration and quenching technique, followed by the electron probe X-ray microanalysis (EPMA) of phase compositions. The liquidus surface of the “$CuO_{0.5}$”-CaO-SiO_2 system has been constructed. The dissolution of copper oxide in the tridymite and cristobalite (SiO_2) was shown to be negligible. The data obtained are used for the optimization of the thermodynamic database for the copper-containing systems.

1. Introduction

In copper smelting, converting and refining processes, refractory materials are exposed to chemically aggressive complex slags; there is a strong financial incentive to increasing the service life of the refractory linings of reactors because they are integral to extending the reactor’s campaign life. A range of slag compositions are used in industrial practice; the elements Cu–Fe–Si–Ca–Al–Mg–O form the basis of the various copper smelting slags in use by industry today [1]. In copper converting and copper recycling technologies, copper metal coexists with fayalite (FeO_x-SiO_2) or calcium ferrite (CaO-FeO_x) based slags [2]. Chrome-magnesite refractories are typically used to contain these materials [3]. It has been shown that as the slags penetrate the pores of the refractories, the iron and silicon components of the slags react with magnesia to form spinel, magnesio-wustite and olivine (Mg_2SiO_4) phases [3–6]. This means that the slag composition within the refractory material differs significantly from the bulk slag. Understanding the phases formed, the proportions and compositions of the phases present at different temperatures within the refractory can assist in improving refractory life and smelter operation.

The aim of the present study is to provide a clearer understanding of the chemical properties of these melts through the characterisation of the “$CuO_{0.5}$”-CaO-SiO_2 system in equilibrium with metallic copper. The “$CuO_{0.5}$”-CaO-SiO_2 system is important for both primary copper production and its pyrometallurgical recycling processes: silica (SiO_2) can be found in copper concentrates and is the main flux in slags used in the pyrometallurgical production of copper [2]; the presence of lime (CaO) in the melts decreases the concentration of copper oxide in slags [7]. The present study forms an important part of the integrated experimental and modelling research program on the Cu–Pb–Zn–Fe–Si–Ca–Al–Mg–O system, aimed at describing the complex chemical systems present in non-ferrous pyrometallurgical primary production and recycling slags. The new experimental data on this important subsystem will enable the subsystems and larger multicomponent systems to be described using chemical thermodynamic models and databases. Phase equilibria data and other thermodynamic properties of these systems will then provide an important foundation for the improved understanding of the interactions of these slags with refractory materials and understanding of ways to improve refractory performance.

The internal thermodynamic database [8] is being developed to thermodynamically describe the equilibria among the most common elements present in metallurgical melts; it includes the following elements, Al–Ca–Cu–Fe–Mg–O–Pb–S–Si–Zn-(Ag, As, Au, Bi, Cr, Ni, Sb, Sn) and includes the description of more than 150 sulphide, sulphate and oxide solid and liquid solutions, and compounds and more than 100 gas species. The database is constantly being reviewed and is periodically

* Corresponding author.
E-mail address: g.khartcyzov@uq.edu.au (G. Khartcyzov).

https://doi.org/10.1016/j.ceramint.2021.12.197
Received 30 August 2021; Received in revised form 16 December 2021; Accepted 17 December 2021
Available online 20 December 2021

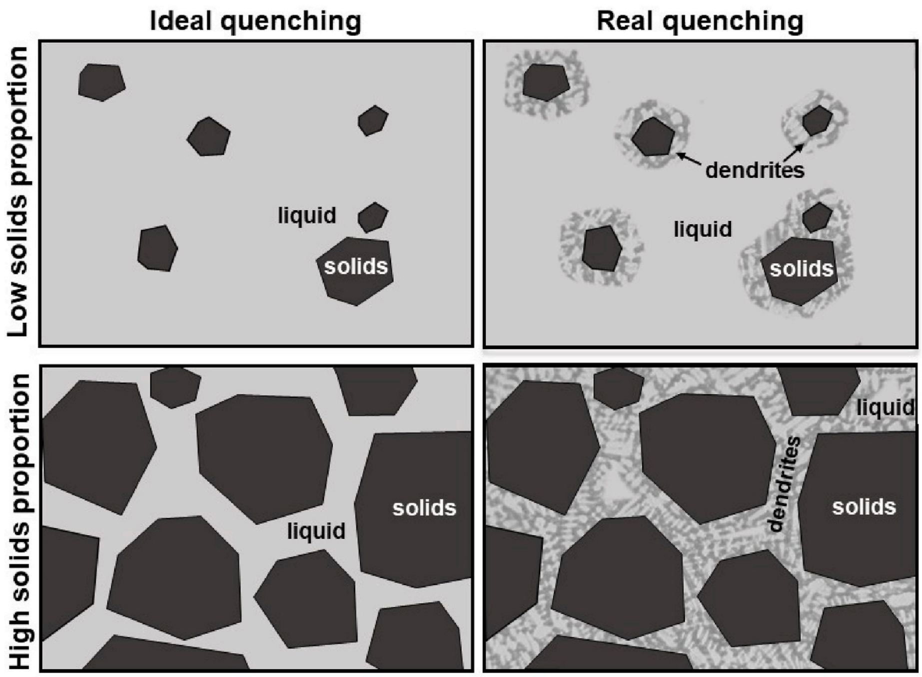

Fig. 1. A schematic diagram illustrating the correlation between the volume fraction of solid phases and the volume fraction of homogenous liquid phase.

updated to achieve a self-consistent set of parameters for the solutions and compounds in the system. This then allows the accurate prediction the phase equilibria and other thermodynamic properties over the wide range of compositions, temperatures and pressures.

The oxidation state of Cu can continuously change from 0 to 2 above 1400 °C in the Cu–O system [9]. Presence of metallic phase makes the contribution of Cu^{2+} negligible, while CaO and SiO_2 supress the formation of Cu^0 in slag, which was also shown by this study. Therefore, quotation marks for "$CuO_{0.5}$" in slag are used in the present study to indicate possible but negligible deviation from Cu^+ towards either Cu^0 or Cu^{2+}. The Cu^+ as a predominant species was confirmed by slag-metal distribution studies as a function of pO_2 [10–12].

The phase equilibria of the "$CuO_{0.5}$"-SiO_2 [13–18] and the "$CuO_{0.5}$"-CaO [13,19–21] systems in equilibrium with metallic copper have been characterised by experimental studies; no binary compounds were found in these pseudo-binary systems. The CaO–SiO_2 system, in addition to the endmembers, lime (CaO) and quartz/-tridymite/cristobalite (SiO_2), contains wollastonite and pseudowollastonite ($CaSiO_3$), rankinite ($Ca_3Si_2O_7$), α'-, α- and γ-dicalcium silicate (Ca_2SiO_4), and tricalcium silicate (Ca_3SiO_5) [22]. Important experimental measurements of the liquidus in the α-dicalcium silicate (Ca_2SiO_4), pseudowollastonite ($CaSiO_3$) and cristobalite (SiO_2) primary phase fields were recently conducted by Cheng et al. [23].

Despite the importance of the "$CuO_{0.5}$"-CaO-SiO_2 system to the metallurgical processing only two experimental studies of this system have been reported to date [15,16]. Liquidus of the ternary system is dominated by the tridymite/cristobalite (SiO_2), pseudowollastonite ($CaSiO_3$) and lime (CaO) primary phase fields. The primary phase fields of other calcium silicates (rankinite ($Ca_3Si_2O_7$), dicalcium silicate (Ca_2SiO_4) and tricalcium silicate (Ca_3SiO_5)) and cuprite (Cu_2O) are comparably smaller. In addition, two ranges of slag-liquid immiscibility were found: one in the high-silica region and one in the low-silica region of the diagram. Hidayat et al. [15] measured liquidus isotherms at 1200 and 1300 °C for the tridymite (SiO_2) and pseudowollastonite ($CaSiO_3$) primary phase fields. The liquid miscibility gap in the low-silica region of the diagram was found by Liu [16] to extend from copper-oxide rich melts, containing less than 2.5 mol.% (2 wt%) SiO_2, to silica concentrations of ~41 mol.% (~40 wt%) SiO_2. No ternary compounds were observed in the system. The thermodynamic optimization of the "$CuO_{0.5}$"-CaO-SiO_2 system and the liquidus surface of this ternary system in equilibrium with metallic copper were reported by Hidayat and Jak [24]. The thermodynamic database developed by Hidayat and Jak [24] is preferred to the public database available in the FactSage 7.3 [25] since the former takes into account more recent experimental data on this system, in particular, it predicts the presence of the liquid immiscibility at low silica compositions, which the public database does not.

In summary, previous researchers studied the region of the diagram bounded with CaO/SiO_2 < 1 join to a maximum temperature of 1350 °C. The present study is focused on the experimental measurement of phase equilibria between liquid, tridymite and cristobalite (SiO_2), cuprite (Cu_2O), pseudowollastonite ($CaSiO_3$), rankinite ($Ca_3Si_2O_7$) and dicalcium silicate (Ca_2SiO_4) in the temperature range between 1160 °C and 1706 °C. The data will be used to re-optimise the parameters in the thermodynamic database [8] and to predict the complete liquidus surface of the "$CuO_{0.5}$"-CaO-SiO_2 system in equilibrium with copper.

2. Experimental technique

The experimental procedure and apparatus were described in previous publications by the authors [24,26–28]. The initial mixtures were made from high-purity powders Cu_2O (>99.9 wt% purity), SiO_2 (>99.9 wt% purity), supplied by Alfa Aesar, MA, USA or Heysham, Great Britain, and CaO–SiO_2 master slags (4:6 and 11:9 M ratio), prepared by high-temperature sintering of mixed SiO_2 and $CaCO_3$ (99.9–99.99 wt% purity; supplied by Alfa Aesar, MA, USA) powders (to promote the formation of calcium silicates and decrease the possible equilibration time required for further experiments). Cu powder (99.9 wt% purity; supplied by Alfa Aesar, MA, USA) was added by 10–20 wt% in excess of the oxide mixture mass to ensure that metallic copper was present at the final equilibrium samples. Less than 0.3 g of mixture was used in each equilibration experiment. The initial compositions of the mixtures were selected so that one or more crystalline phases were present in equilibrium with liquid slag. The volume fraction of solids was targeted to be below 50%, and preferably about 10%, to achieve rapid equilibration and satisfactory quenching of the liquid phase to amorphous material. It was found that the solid phases acted as heterogeneous nucleation centres, and a minimum distance of 10–20 μm between solid grains was necessary to ensure that an amorphous slag of uniform composition, unaffected by dendritic crystal growth during quenching, was obtained. The correlation between the volume fraction of the solid phases and the volume fraction of the slag phase affected by the growth of the quenching crystals is schematically shown in Fig. 1: an increase in the volume of the solid phases will lead to a decrease in the volume of the slag suitable for accurate measurement using the EPMA. An iterative procedure involving preliminary experiments was often required to achieve the targeted phase proportions for a given equilibration temperature and composition, since the exact liquidus compositions at a given temperature were not initially known. Only homogenous slag areas, that is, those not affected by the growth of the "quench crystals", were analysed and included in the present study.

The mixtures were pressed into pellets using a tool steel die (8 mm in diameter) before being mounted on a substrate. Three types of sample support substrates were used for equilibration, depending on the target compositions:

1. Silica ampoules (5 cm long, 1.3 cm outer diameter, 99.999%, supplied by GY, Jiangsu, China) were used for the experiments in the tridymite/cristobalite (SiO_2) primary phase fields;
2. $CaSiO_3$ crucibles sintered from pure $CaCO_3$ and SiO_2 at 1470 °C (for 16 h in air) were used for the pseudowollastonite ($CaSiO_3$) primary phase field experiments; the synthesized $CaSiO_3$ was found to be stoichiometric within the electron probe X-ray microanalysis uncertainty (50.2 ± 0.1 mol.% CaO, 49.8 ± 0.1 mol.% SiO_2, and <0.1 mol.% in other elements);
3. Iridium wire (99.99%, supplied by Huanya, Jiangsu, China) substrate, 50–60 mm long, wound into a spiral shape, was used for the investigation of the pseudowollastonite ($CaSiO_3$), rankinite ($Ca_3Si_2O_7$) and dicalcium silicate (Ca_2SiO_4) primary phase fields. This type of substrate support was chosen for several reasons: 1) Ir does not react with slag components and reacts limitedly with metallic copper; and 2) the synthesis of a homogenous rankinite ($Ca_3Si_2O_7$) or dicalcium silicate (Ca_2SiO_4) substrate is a difficult procedure. Volume change associated with the polymorphic transformation of dicalcium silicate (Ca_2SiO_4) on cooling below 800 °C leads to the disintegration of the substrate. The use of the iridium over platinum was preferred mainly based on its higher corrosion resistance to copper which prevents the destruction of the sample during the equilibration experiment. The use of Pt or Ir changes the phase equilibria from pure Cu metal to Cu–Pt or Cu–Ir alloy. For Pt, this would mean a significant decrease of activity of Cu (~0.7, according to the Cu–Pt phase diagram [29] and assuming Raoultian behaviour.) and lead to the destruction of the Pt support. For Ir, the decrease of Cu activity is less significant (~0.90, according to the Cu-Ir [30] phase diagram and assuming Raoultian behaviour; only 0.1–0.2 mol% Ir was measured in the copper phase for the experiments where Ir support was used) which was confirmed by direct measurements of the metal composition. At the same time, the evaporation of Cu sets an upper temperature limit for the use of an open substrate to 1500 °C.

Each equilibration experiment was carried out in an electrical, resistance-heated vertical tube furnace with an impervious recrystallized alumina (30-mm inner diameter) reaction tube. The heating elements in the furnaces were made of silicon carbide (SiC). The samples were placed immediately adjacent to a working thermocouple (type B), that was encased in a recrystallized alumina sheath in the uniform hot zone of the furnace, to monitor the actual sample temperature. The working thermocouple was calibrated against a standard thermocouple (supplied by the National Measurement Institute of Australia, NSW, Australia) and the melting point of cristobalite (1723 °C). The overall absolute temperature accuracy of the experiment was estimated to be ±3 °C.

The samples were suspended in the uniform hot zone of the furnace by Kanthal support wire (Fe–Cr–Al alloy, 0.7 or 1-mm diameter, supplied by Vulcan Stainless, VIC, Australia). For temperatures higher than 1500 °C, when Kanthal support wire can no longer be used (melting point of alloy is specified by the manufacturer as 1500 °C), a 20–25 cm long Pt or Pt–Rh wire (99.99 wt%, 0.5 mm diameter, supplied by Johnson Matthey Ltd, VIC, Australia) was used to extend the support, so that the Kanthal wire is not in the hot zone. For selected experiments, the samples were preheated for 15–60 min at 20–50 °C above the equilibration temperature, to form a homogeneous liquid slag, and/or precooled, to promote the formation of multiple solid phases rather than supercooled liquid. The formation of the homogenous liquid slag at the preheating temperature was predicted for every composition using Factsage 7.3 [25] and internal thermodynamic database [8]. This was followed by the equilibration at the final target temperature for the required time. In individual case the constant $p(O_2) = 10^{-11.5}$ atm was maintained by an accurate control of the CO and CO_2 ratios in the gas phase using the system of calibrated U-tube capillary flow-meters. The desired flowrates of gases, corresponding to the indicated $p(O_2)$ were calculated using FactSage 7.3 FactPS database [25] for the ideal gas phase. The accuracy of the oxygen potential was confirmed using a DS-type oxygen probe (supplied and calibrated by Australian Oxygen Fabricators, AOF, Melbourne, Australia). At the end of the equilibration process, the samples were released and rapidly quenched into the $CaCl_2$ brine solution (prepared from DampRid Moisture Absorber and stored at −20 °C in the freezer). Brine solution was chosen over water due to the higher cooling rates [31,32]. The samples were then washed thoroughly in water and ethanol, dried on a hot plate, mounted in epoxy resin and cross-sectioned and polished using conventional metallographic techniques.

The cross-sectioned samples were examined by optical microscopy, carbon-coated, and the compositions of the phases were measured by Electron Probe X-ray Microanalysis (EPMA) with the wavelength-dispersive detectors (JEOL 8200L EPMA; Japan Electron Optics Ltd., Tokyo, Japan). The EPMA was operated with 15 kV accelerating voltage and 20 nA probe current using the standard Duncumb–Philibert atomic number, absorption, and fluorescence (ZAF) correction supplied by JEOL [33–35]. Copper (Cu) (for Cu and "$CuO_{0.5}$" measurements), corundum (Al_2O_3) (for $AlO_{1.5}$ measurements), hematite (Fe_2O_3) (for $FeO_{1.5}$ measurements) and wollastonite ($CaSiO_3$) (for CaO and SiO_2 measurements) (supplied by Charles M. Taylor Co., Stanford, CA) were used as the EPMA standards. The concentrations of metal cations were measured with EPMA; no data on the oxidation states of the metal cations in slag were obtained. The concentrations of aluminium oxide and iron oxide were also measured to ensure that samples were not contaminated with Kanthal wire dust or Al_2O_3 furnace tubes. For presentation purposes, the copper oxide concentrations were then re-calculated as "$CuO_{0.5}$". It was shown [36] that the increase from fully focused electron beam (zero probe diameter) to 50 μm decreased the uncertainty of the copper slags compositions measured with EPMA. The probe diameter was selected depending on the size of the phase under investigation. The slag compositions reported in this study were measured mainly using a 50 μm probe diameter. The compositions of the phases reported in the present study are the mean values of a number of independent measurements from different regions of the samples rather than single measurements.

For the purpose of the tridymite vs. cristobalite (SiO_2) identification, the samples were designed to contain >50% solids and were not then suitable for the accurate EPMA measurements of the liquid slag. The samples were preheated up to 1530 °C for 10 min and then equilibrated for 16 h. When the equilibration process was finished, the samples were quenched into $CaCl_2$ brine solution (−20 °C), washed in water and ethanol and dried on the hot plate. A piece of each sample was then taken for the following electron probe X-ray microanalysis (EPMA) and the rest of it was crushed into a powder using agate mortar and pestle ad

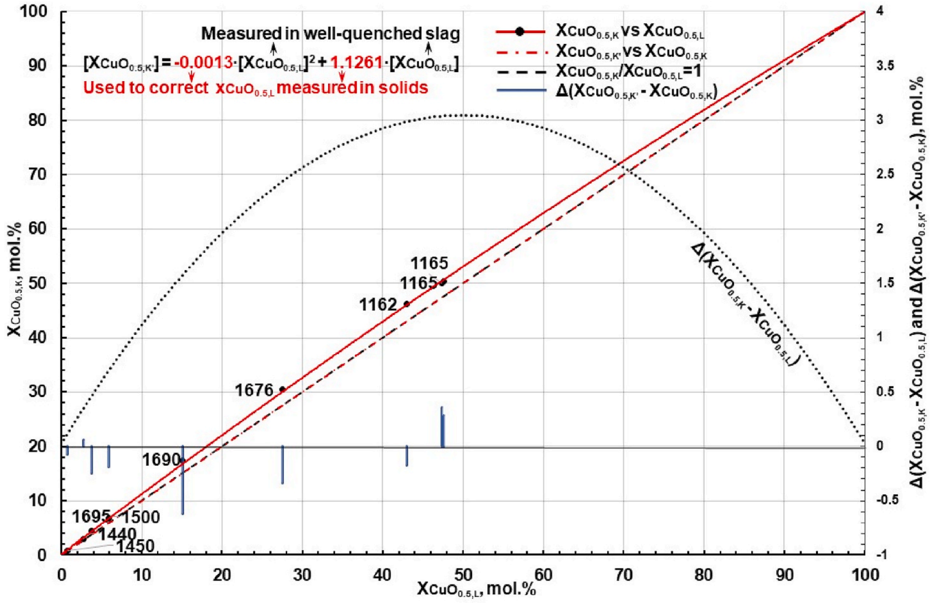

Fig. 2. Analytical correction formula used in the present study to take into the account the secondary X-ray fluorescence in copper containing materials. $x_{CuO0.5,K}$ and $x_{CuO0.5,L}$ are the Cu oxide concentration values measured with EPMA using K_α-values and L_α-values respectively after the standard JEOL ZAF correction in mol.%, and $x_{CuO0.5,K'}$ are the apparent "pseudo" K_α Cu oxide concentration values derived from $x_{CuO0.5,L}$ also in mol.%.; temperatures in equilibrium are given in °C.

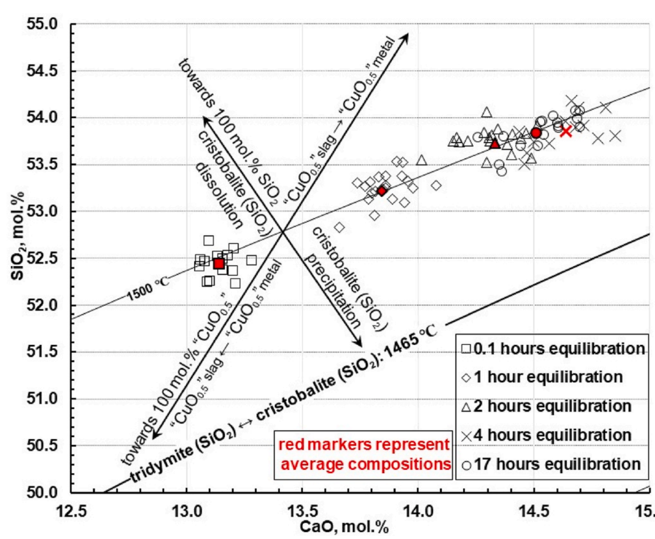

Fig. 3. The experimental correlation between the composition of liquid phase and equilibration time in the "$CuO_{0.5}$"-CaO-SiO_2 system in equilibrium with metallic copper at 1500 °C.**2.10**.

studied with X-ray powder diffraction (XRPD).

The phase assemblage of selected samples was confirmed using Bruker D8 Advance X-ray diffractometer with a LynxEye detector operating with Cu K radiation (λ = 1.5406 Å) at 40 kV and 40 mA. Data were collected in a range from 15° to 70° with a speed of $0.5°\cdot min^{-1}$. The phases present were identified from XRD peaks using DIFFRAC.EVA V5.1 software and PDF-4+ 2021 database by the International Centre for Diffraction Data.

2.1. Custom correction of the standard JEOL ZAF correction

The accuracy of the compositional measurements was improved on that obtained using the standard JEOL ZAF correction by following an approach similar to that described previously by the authors [18, 37–39]. This involves using stoichiometric compounds of known composition as secondary internal standards. The SiO_2 concentration in the stoichiometric dehydrated dioptase ($CuSiO_3$) measured with EPMA was found to be overestimated by 1.8 mol.% [18] after the standard JEOL ZAF correction. The CaO concentration in the stoichiometric Ca_2CuO_3 phase (synthesized in the present study from the $CaCO_3$ + CuO mixture in air on the 99.99% Au foil at 1020 °C with a small excess of CuO to ensure the stoichiometric Ca_2CuO_3 crystals are in equilibrium with the "CuO"-CaO slag) measured with EPMA after the standard JEOL ZAF correction was found to be overestimated by 0.78 mol.% Ca/(Ca+Cu) above its stoichiometric value (66.67 mol.% CaO). These values were used to derive the following custom correction formulae, which were used in this study:

$$x_{CuO_x}^{corrected} = x_{CuO_x} + 0.072 \times x_{SiO_2}x_{CuO_x} + 0.035 \times x_{CaO}x_{CuO_x} \tag{1}$$

$$x_{CaO}^{corrected} = x_{CaO} - 0.035 \times x_{CaO}x_{CuO_x} \tag{2}$$

$$x_{SiO_2}^{corrected} = x_{SiO_2} - 0.072 \times x_{SiO_2}x_{CuO_x} \tag{3}$$

where x(CuO_x, CaO, SiO_2) are the cation molar fractions after the standard JEOL ZAF corrections, and x(CuO_x, CaO, SiO_2)^{corrected} are the corrected cation molar fractions respectively.

2.2. Custom correction for the effect of secondary X-ray fluorescence

The overestimation of concentrations of some elements, such as Fe, Cu, Zn in tridymite or cristobalite (SiO_2) within the Fe, Cu, Zn containing matrix (slag) measured with EPMA, due to the secondary fluorescence by the X-rays originated from the continuum background spectra of the inelastic scattering of electrons (bremsstrahlung), was previously reported [40,41]. The same effect was observed by the authors in previous studies (Fe [42,43], Cu [18,44], Zn [43–45]). By undertaking EPMA measurements of these elements in unreacted pellets consisted of the 10–100 μm pure SiO_2 particles mechanically pressed within the Fe_2O_3, Cu or ZnO powder matrix, the concentrations of these elements, reported to be solid solution in tridymite and cristobalite (SiO_2) in the slag containing samples, were proven to be overestimated values. This bremsstrahlung effect was only detected for the measurements using characteristic X-ray lines with energies above 5 keV; the X-rays from the

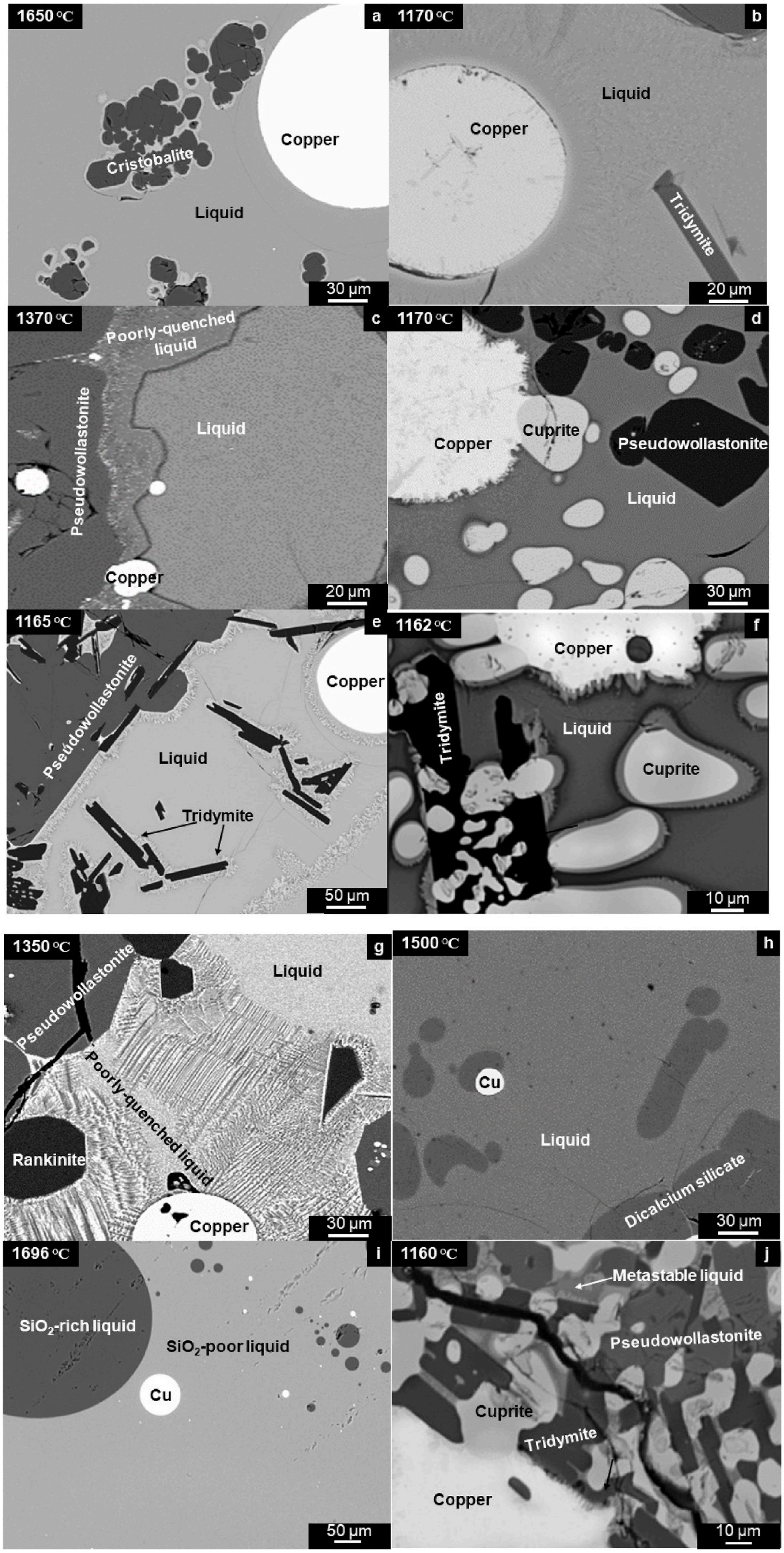

Fig. 4. Backscattered SEM micrographs of the quenched "$CuO_{0.5}$"-CaO-SiO_2 samples in equilibrium with metallic copper: (a) liquid-cristobalite(SiO_2)-metal; (b) liquid-tridymite(SiO_2)-metal; (c) liquid-pseudowollastonite($CaSiO_3$)-metal; (d) liquid-pseudowollastonite($CaSiO_3$)-cuprite(Cu_2O)-metal; (e) liquid-tridymite(SiO_2)-pseudowollastonite($CaSiO_3$)-metal; (f) liquid -tridymite(SiO_2)-cuprite(Cu_2O)-metal; (g) liquid-rankinite($Ca_3Si_2O_7$)-pseudowollastonite($CaSiO_3$)-metal; (h) liquid-rankinite($Ca_3Si_2O_7$)-dicalcium silicate(Ca_2SiO_4)-metal; (i) 2 liquids-metal describing immiscibility gap in high-silica area of the system; (j) fully solid system at 1160 °C representing subsolidus condition for the pseudowollastonite($CaSiO_3$)-cuprite(Cu_2O)-tridymite(SiO_2).

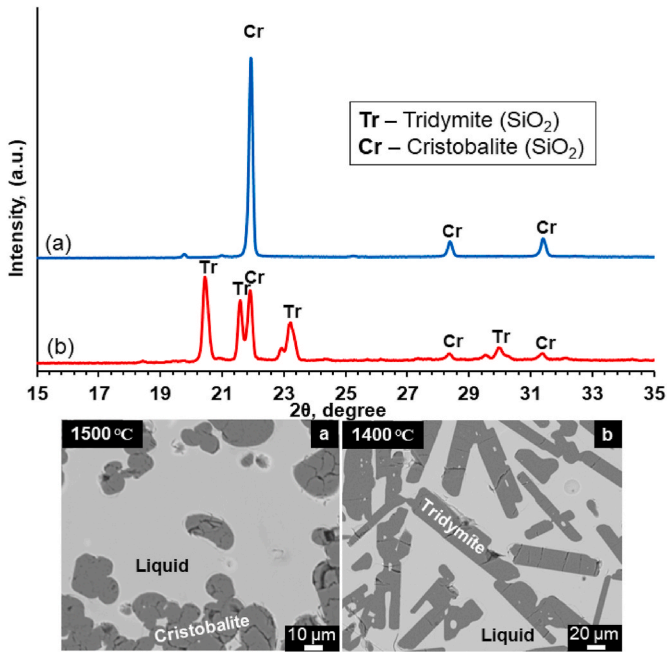

Fig. 5. XRD patterns and microstructures for the samples equilibrated at different temperatures.

Table 1
Experimental measurements of phase compositions of the XRPD samples.

Code	Temperature, °C	Phases present	Compositions (mol.%)		
			"$CuO_{0.5}$"	CaO	SiO_2
a	1500	Liquid	16.7 ± 0.3	22.5 ± 0.3	60.8 ± 0.5
		Cristobalite	<0.01	<0.1	>99.9
b	1400	Liquid	27.9 ± 0.3	17.2 ± 0.1	55.0 ± 0.2
		Tridymite	<0.01	<0.1	>99.9

continuum background with these energies have high (>100 μm) penetration ability through the SiO_2 or other particles of comparable density into the surrounding matrix (e.g. slag) thus exciting the characteristic lines of Fe, Cu, Zn and therefore generating the incorrect apparent overestimated values of concentrations of these elements in such particles [40,41]. Measurements using the characteristic X-ray lines with low energy do not have this effect due to the strong absorption of that spectrum range of the background X-rays [40,41]. The solubilities of Cu oxide in tridymite and cristobalite (SiO_2) measured in the present study using L_α-line (0.93 keV) were found to be close to zero (0–0.05 mol.%) (compared to the apparent values of ~1 mol.% in case of the use of the K_α-line with 8.05 keV); these observations confirmed that the direct measurement of the Cu concentration in the low-Cu-oxide phases with the L_α-line (0.93 keV) gives improved accuracy.

Preliminary measurements of the Cu oxide concentrations in phases with known compositions using Lα-line (0.93 keV) indicated significant uncertainty in the corresponding standard JEOL ZAF corrections. The custom additional correction for the concentrations measured with the K_α-lines (equations (1)–(3)) had already been developed and tested by the authors [18,37–39]. The following procedure therefore was developed and used in the present study for measurements of the Cu concentration in the low-Cu-oxide phases. The copper oxide concentrations were measured in the same areas of several well-quenched slag samples using both Cu K_α and L_α lines. The values obtained were found to be different from each other with (Cu K_α)/(Cu L_α) ratio being >1 for the whole measured range of concentrations (Fig. 2). The following correlation was developed and used to re-calculate the concentrations of copper oxide measured with L_α lines into correct "K" concentrations:

$$x_{CuO0.5,K'} = -0.0013\ (x_{CuO0.5,L})^2 + 1.1261\ x_{CuO0.5,L} \quad (4)$$

where $x_{CuO0.5,L}$ are the Cu oxide concentration values measured with EPMA using L_α-values after the standard JEOL ZAF correction in mol.%, and $x_{CuO0.5,K'}$ are the apparent "pseudo" K_α Cu oxide concentration values derived from $x_{CuO0.5,L}$ also in mol.%. equations (1)–(3) were then applied using the $x_{CuO0.5,K'}$ values derived from $x_{CuO0.5,L}$.

2.3. Confirmation of achievement of equilibrium

To ensure the achievement of equilibrium in the samples, the "4-point test" [28,46] was used including: (1) variation of equilibration time; (2) assessment of the compositional homogeneity of the phases by EPMA; (3) approaching the final equilibrium point from different starting conditions; and, importantly, (4) consideration of reactions specific to this system that may affect the achievement of equilibrium or reduce the accuracy.

To comply with these principles, additional experiments were carried out during this study. As an example of this procedure, 5 samples of the same initial composition (30 mol.% "$CuO_{0.5}$", 10 mol.% CaO and 60 mol.% SiO_2) were sealed in SiO_2 ampoules and equilibrated at 1500 °C for 0.1, 1, 2, 4 and 17 h, quenched and studied with EPMA to establish the time required to achieve the equilibrium. The results of the measurements after corrections are shown in Fig. 3. The symbols in red are the mean values obtained after each of the respective reaction times. The compositions of liquid phases for samples held for 2, 4 and 17 h were found to be in agreement with each other and different from those held for 0.1 and 1 h. The difference between the initial and final compositions of the samples was explained by two reactions which influence the composition of the liquid phase in every primary phase field of the diagram: 1) the dissolution/precipitation of primary phase (cristobalite (SiO_2) for the experiment described), and 2) the "$CuO_{0.5}$" exchange between the slag phase and metallic copper. As shown in Fig. 3, these reactions can change the composition of the liquid phase in two directions – towards or away from SiO_2 and "$CuO_{0.5}$" corners of the diagram respectively. The change of the composition towards SiO_2 corner of the diagram will indicate the ongoing dissolution of either SiO_2 substrate or precipitated SiO_2 solids, away from SiO_2 – precipitation of the SiO_2 solids. The change of the composition towards "$CuO_{0.5}$" corner of the diagram will indicate the transfer of copper oxide from the metallic phase to the liquid slag, away from "$CuO_{0.5}$" – transfer of copper oxide from the liquid phase to the metallic phase. The final change in the composition should then be the combination of these reactions in order for the liquid to stay on the corresponding isotherm. The direction of compositional change of the liquid with time indicates the nett precipitation of cristobalite (SiO_2) from the liquid and the transfer of "$CuO_{0.5}$" from liquid phase to metallic copper. The precipitation of cristobalite (SiO_2) with time indicates the completeness of the dissolution of the raw material in the liquid within the first minutes after reaching the equilibration temperature. The average liquid compositions held for 2, 4 and 17 h are the same within the standard deviations of the measurements. The minimum equilibration time depends on the phases present, bulk composition and temperature.

Only well-quenched areas of the liquid were chosen for the analysis and the accuracies of the analyses were demonstrated in each case by low standard deviations. For some samples, different initial compositions, heat-treatment procedures (preheat/precool, cooling rate and the equilibration time) or types of substrate were used to ensure the achievement of the equilibrium on the boundary lines, prevent the formation of possible metastable phases or minimize the influence of ongoing dissolution/precipitation reactions on the equilibration process. Only after these procedures and the final methodology were established, the compositions of phases measured were accepted as those

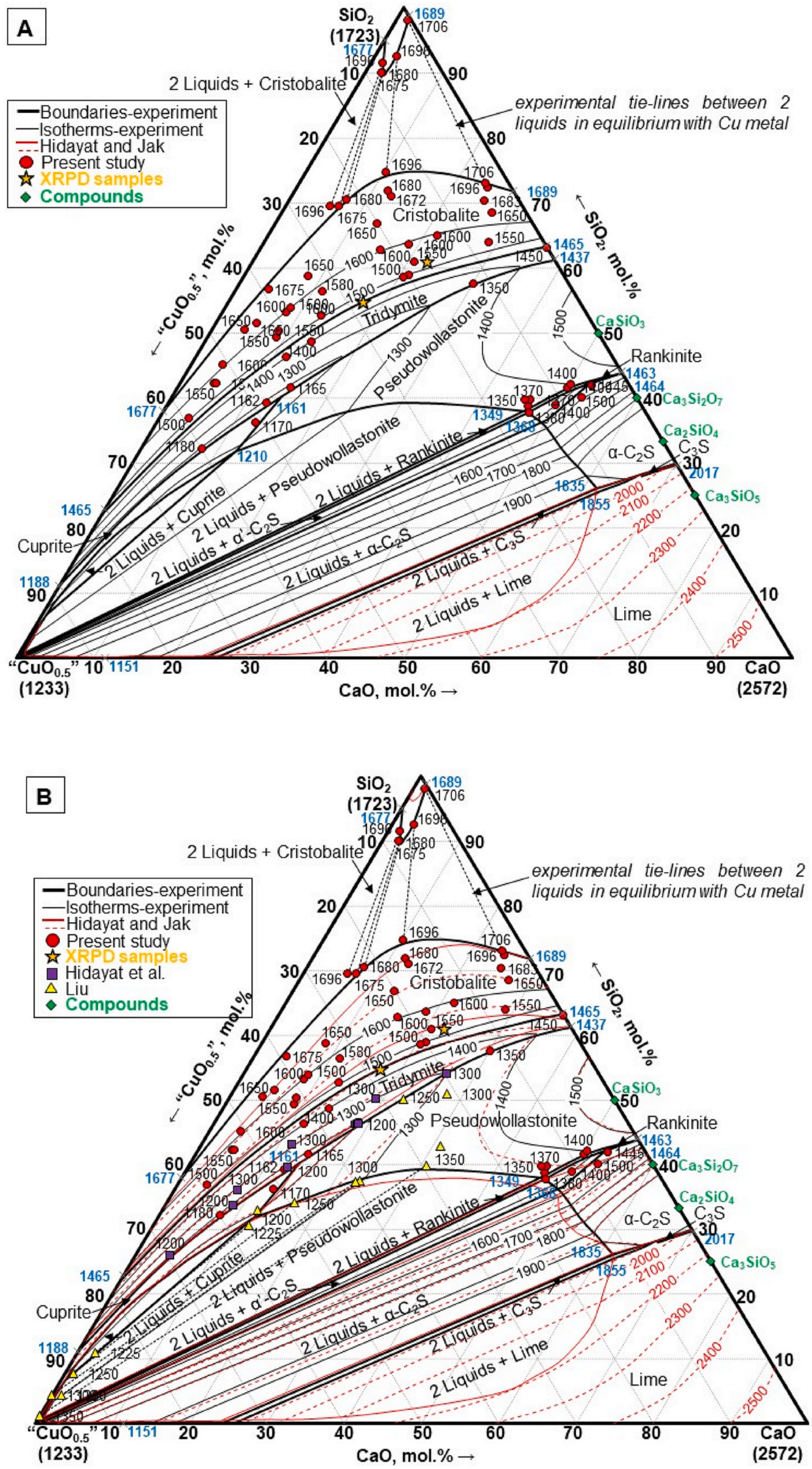

Fig. 6. Liquidus surface of the "$CuO_{0.5}$"-CaO-SiO_2 system in equilibrium with metallic copper. Phase boundaries and isotherms at CaO/SiO_2 < 60/40 mol.% were constructed from the experimental results, at CaO/SiO_2 > 60/40 mol.% using model predictions by Hidayat and Jak [24]. (a) The experimental results of present study; (b) The comparison of experimental results with data from Refs. [15,16,24].Temperatures are given in °C.

Table 2
Experimental measurements of phase compositions for the "$CuO_{0.5}$"-CaO-SiO_2 system in equilibrium with metallic copper.

Substrate	Preheat/precool	Final equilibration	Time	Phases present	Compositions (mol.%)			Calculated log p(O_2)
	°C		h		"$CuO_{0.5}$"	CaO	SiO_2	
Tridymite/Cristobalite (SiO_2) liquidus								
SiO_2	1190/1150	1160	14	Tridymite	0.00	<0.03	>99.97	–
				Cuprite	>99.9	<0.1	<0.01	
				Pseudowollastonite	<0.07	50.0	50.0	
SiO_2	1190/1150	1162	14	Liquid	48.2 ± 0.1	12.7 ± 0.1	39.1 ± 0.1	−4.5
				Tridymite	0.00[L]	0.03	99.97	
				Cuprite	99.92	0.08	0.00	
SiO_2	1190/1150	1165	4	Liquid	43.9 ± 0.3	14.6 ± 0.2	41.5 ± 0.2	−4.6
				Tridymite	0.03[L]	0.04	99.3	
				Pseudowollastonite	0.09[L]	50.4	49.5	
SiO_2	1200/1160	1180	3	Liquid	59.9 ± 0.7	8.0 ± 0.3	32.1 ± 0.4	−4.4
				Tridymite	0.00	<0.03	>99.97	
SiO_2	1390/1320	1350	4	Liquid	12.3 ± 0.2	30.3 ± 0.1	57.4 ± 0.1	−5.2
				Tridymite	0.00	<0.03	>99.97	
				Pseudowollastonite	<0.07	50.0	50.0	
SiO_2	1430/-	1400	16	Liquid	42.0 ± 0.2	11.6 ± 0.1	46.3 ± 0.1	−3.6
				Tridymite	<0.01	<0.1	>99.9	
SiO_2	1430/-	1400	16	Liquid	37.4 ± 0.2	13.9 ± 0.1	48.6 ± 0.2	−3.8
				Tridymite	<0.01	<0.1	>99.9	
SiO_2	1430/-	1400	3	Liquid	37.6 ± 0.3	13.8 ± 0.1	48.6 ± 0.3	−3.8
				Tridymite	0.00[L]	<0.03	>99.97	
SiO_2	−/−	1450	4	Liquid	0.07 ± 0.03	37.0 ± 0.2	63.0 ± 0.2	−11.5[1]
				Tridymite	<0.01	<0.1	>99.9	
SiO_2	1530/-	1500	1	Liquid	19.9 ± 0.1	21.2 ± 0.1	58.9 ± 0.1	−4.3
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	1530/-	1500	16	Liquid	20.4 ± 0.5	20.9 ± 0.2	58.6 ± 0.7	−4.3
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	−/−	1500	17	Liquid	31.7 ± 0.3	14.5 ± 0.1	53.8 ± 0.2	−3.7
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	1530/-	1500	1	Liquid	34.4 ± 0.5	13.0 ± 0.3	52.6 ± 0.3	−3.5
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	1530/-	1500	1	Liquid	59.3 ± 0.7	3.9 ± 0.1	36.8 ± 0.6	−3.0
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	1580/-	1550	1	Liquid	7.2 ± 0.1	29.0 ± 0.1	63.8 ± 0.1	–5.7
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	1580/-	1550	1	Liquid	18.2 ± 0.2	20.9 ± 0.1	60.9 ± 0.2	−4.3
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	1580/-	1550	1	Liquid	40.9 ± 0.3	8.8 ± 0.1	50.3 ± 0.3	−3.1
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	1580/-	1550	2	Liquid	41.8 ± 0.2	8.9 ± 0.1	49.3 ± 0.2	−3.1
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	1580/-	1550	1	Liquid	53.2 ± 0.4	4.5 ± 0.1	42.3 ± 0.4	−2.8
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	1580/-	1550	2	Liquid	53.1 ± 0.4	4.8 ± 0.1	42.1 ± 0.3	−2.8
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	1600/-	1580	1	Liquid	32.3 ± 0.2	11.4 ± 0.1	56.3 ± 0.2	−3.3
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	1620/-	1600	1	Liquid	13.3 ± 0.1	21.8 ± 0.1	64.9 ± 0.1	−4.6
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	1620/-	1600	1	Liquid	21.6 ± 0.2	15.7 ± 0.1	62.7 ± 0.2	−3.8
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	1620/-	1600	1	Liquid	37.7 ± 0.3	8.6 ± 0.1	53.7 ± 0.3	−3.0
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	1620/-	1600	2	Liquid	38.5 ± 0.3	8.3 ± 0.1	53.2 ± 0.2	−3.0
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	1620/-	1600	1	Liquid	17.6 ± 0.5	18.9 ± 0.3	63.5 ± 0.2	−4.2
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	1620/-	1600	1	Liquid	14.9 ± 0.6	21.6 ± 0.5	63.5 ± 0.3	−4.5
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	1620/-	1600	1	Liquid	50.8 ± 0.6	4.2 ± 0.2	45.0 ± 0.6	−2.7
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	−/−	1650	1	Liquid	4.4 ± 0.2	27.1 ± 0.2	68.5 ± 0.2	−6.4
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	−/−	1650	0.5	Liquid	20.0 ± 0.1	13.2 ± 0.1	66.8 ± 0.1	−3.6
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	−/−	1650	1	Liquid	32.9 ± 0.2	8.4 ± 0.1	58.7 ± 0.3	−2.9
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	−/−	1650	1	Liquid	45.2 ± 0.5	4.4 ± 0.1	50.4 ± 0.4	−2.6
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	−/−	1650	1	Liquid	43.2 ± 0.5	5.3 ± 0.1	51.5 ± 0.5	−2.7
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	−/−	1672	0.5	Liquid	11.0 ± 0.2	16.1 ± 0.2	72.9 ± 0.2	−4.6
				Cristobalite	<0.01	<0.1	>99.9	

(continued on next page)

Table 2 (*continued*)

Substrate	Preheat/precool	Final equilibration	Time	Phases present	Compositions (mol.%)			Calculated log $p(O_2)$
	°C		h		"$CuO_{0.5}$"	CaO	SiO_2	
SiO_2	−/−	1675	1	Liquid	39.0 ± 0.4	4.3 ± 0.2	56.7 ± 0.4	−2.5
				Cristobalite	0.07^{L}	0.08	99.85	
SiO_2	−/−	1680	1	Liquid	16.0 ± 0.3	12.1 ± 0.1	71.9 ± 0.4	−3.8
				Cristobalite	<0.01	<0.1	>99.9	
SiO_2	−/−	1683	1	Liquid	4.4 ± 0.1	25.3 ± 0.2	70.3 ± 0.2	−6.4
				Cristobalite	0.01^{L}	0.12	99.87	
SiO_2	−/−	1696	1	Liquid	3.0 ± 0.2	24.6 ± 0.2	72.4 ± 0.2	−6.4
				Cristobalite	<0.01	<0.1	>99.9	
2 immiscible liquids								
SiO_2	−/−	1675	1	SiO_2-rich liquid	7.9 ± 0.6	2.1 ± 0.2	90.0 ± 0.7	−2.8
				SiO_2-poor liquid	23.7 ± 0.5	6.9 ± 0.2	69.4 ± 0.7	
SiO_2	−/−	1680	1	SiO_2-rich liquid	7.7 ± 0.3	2.3 ± 0.2	90.0 ± 0.4	−2.9
				SiO_2-poor liquid	22.1 ± 0.5	7.4 ± 0.2	70.5 ± 0.6	
SiO_2	−/−	1696	1	SiO_2-rich liquid	6.9 ± 0.4	1.6 ± 0.1	91.5 ± 0.5	−2.9
				SiO_2-poor liquid	24.9 ± 0.7	5.8 ± 0.2	69.3 ± 0.6	
SiO_2	−/−	1696	1	SiO_2-rich liquid	4.7 ± 0.2	2.8 ± 0.1	92.5 ± 0.3	−3.6
				SiO_2-poor liquid	14.9 ± 0.1	10.4 ± 0.1	74.7 ± 0.2	
SiO_2	−/−	1706	1	SiO_2-rich liquid	0.4 ± 0.1	1.5 ± 0.1	98.1 ± 0.2	−7.1
				SiO_2-poor liquid	3.0 ± 0.1	24.0 ± 0.2	73.0 ± 0.2	
Pseudowollastonite ($CaSiO_3$) liquidus								
$CaSiO_3$	1190/1150	1170	4	Liquid	51.0 ± 0.4	12.9 ± 0.3	36.1 ± 0.3	−4.5
				Pseudowollastonite	0.14^{L}	49.8	50.0	
				Cuprite	99.95	0.05	0.0	
Ir wire	1400/-	1370	2	Liquid	13.9 ± 0.2	46.5 ± 0.2	39.6 ± 0.2	−3.4
				Pseudowollastonite	0.07^{L}	50.1	49.8	
$CaSiO_3$	1400/-	1370	3	Liquid	14.6 ± 0.2	45.8 ± 0.1	39.6 ± 0.1	−3.3
				Pseudowollastonite	0.13^{L}	50.1	49.8	
$CaSiO_3$	1430/-	1400	3	Liquid	7.5 ± 0.4	50.4 ± 0.3	42.1 ± 0.3	−4.0
				Pseudowollastonite	0.10^{L}	49.9	50.0	
Rankinite ($Ca_3Si_2O_7$) + Pseudowollastonite ($CaSiO_3$) liquidus								
Ir wire	1390/1330	1350	3	Liquid	14.6 ± 0.2	46.7 ± 0.1	38.7 ± 0.2	−3.3
				Rankinite	0.13^{L}	60.0	39.87	
				Pseudowollastonite wollastonite	<0.07	50.0	50.0	
Ir wire	1420/1380	1400	1	Liquid	8.1 ± 0.3	50.3 ± 0.5	41.6 ± 0.3	−3.9
				Rankinite	0.10^{L}	60.05	39.85	
				Pseudowollastonite	<0.07	50.0	50.0	
Dicalcium silicate (Ca_2SiO_4)								
Ir wire	−/−	1380	0.25	"$CuO_{0.5}$"-rich liquid	98.9 ± 0.4	0.82 ± 0.2	0.24 ± 0.2	−3.3
				"$CuO_{0.5}$"-poor liquid	15.0 ± 0.1	47.4 ± 0.3	37.6 ± 0.2	
				Dicalcium silicate	0.26^{L}	66.7	33.0	
Ir wire	1420/-	1400	1	Liquid	11.1 ± 0.2	50.1 ± 0.2	38.8 ± 0.2	−3.6
				Dicalcium silicate	0.26^{L}	66.9	32.8	
Ir wire	1470/-	1445	1	Liquid	4.9 ± 0.1	53.3 ± 0.1	41.8 ± 0.2	−4.4
				Dicalcium silicate	0.26^{L}	66.3	33.5	
Ir wire	1520/-	1500	0.5	Liquid	7.1 ± 0.1	52.8 ± 0.1	40.1 ± 0.1	−4.0
				Dicalcium silicate	0.90^{L}	65.8	33.3	

[1] Experimental value.

Note: [L] indicates that Cu L_{α}-line was used for measurement, with subsequent correction according to equations (4) and (1-3). For all other cases, Cu Kα-line was used.

representing the equilibrium condition.

2.4. Equilibrium atmosphere

Note that the present study for the equilibrium with copper metal has been undertaken under closed system conditions, that is

the oxygen partial pressures are not fixed but vary with temperature and slag composition. In this situation, the pO_2 is controlled by the $CuO_{0.5}$ activity in slag and Cu activity in metal through the equation,

$$Cu(l) + 0.25O_2(g) \leftrightarrow CuO_{0.5}(l),\ \Delta_f G^0 = -59793 + 21.074T$$

$$\log(pO_2)4\frac{\Delta_f G^0}{2.303RT} + 4\log\left(\frac{a_{CuO_{0.5}}}{a_{Cu}}\right) \tag{5}$$

The solubilities of Ca and Si in liquid copper under these conditions have been found to be very low, below the limits of detection using EPMA. Thermodynamic calculations show that the copper metal in equilibrium with the pure cuprite (Cu_2O) phase at its melting temperature, 1233 °C [9] contains approximately 8 mol.% O. The copper activity in the system for all liquidus conditions studied is therefore very close to unity. For the liquidus of cuprite (Cu_2O) the effective oxygen partial pressures can therefore be readily calculated using the equilibrium relations for equation (5) and knowing the Gibbs free energy of the reaction. For the liquidus of other primary phases, the activity of "$CuO_{0.5}$" should be known. The phase diagrams built in the course of this study are those in equilibrium with metallic copper, representing the lowest equilibrium partial pressure of oxygen. These pressures were not measured in this study and can be estimated using equation (5).

3. Results and discussion

3.1. The "$CuO_{0.5}$"-CaO-SiO_2 liquidus

Examples of typical microstructures observed in the samples quenched from the equilibration temperatures are given in Fig. 4. All micrographs were obtained using backscattered electron microscopy (BSE) to provide contrast based on mean atomic number. All micrographs demonstrated the coexistence of liquid copper metal with liquid slag. The phases illustrated in these figures include cristobalite (SiO_2) (4a), tridymite (SiO_2) (4b, e, j), cuprite (Cu_2O) (4d, f, j), pseudowollastonite ($CaSiO_3$) (4c, d, j), rankinite ($Ca_3Si_2O_7$) (4g), dicalcium

Table 3
Phase equilibria measurements for the "$CuO_{0.5}$"-CaO-SiO_2 system in equilibrium with metallic copper obtained by Liu [16], used for construction of the present phase diagram.

Temperature	Equilibration	Phases present	Compositions (mol.%)		
°C	h		"$CuO_{0.5}$"	CaO	SiO_2
1350	1	"$CuO_{0.5}$"-poor liquid	29.4	30.7	39.9
		*"$CuO_{0.5}$"-rich liquid	98.9	0.00	1.1
		Pseudowollastonite	<0.34	49.6	50.1
1350	1	Liquid	26.1	31.0	42.9
		Pseudowollastonite	<0.86	49.5	49.7
1300	2	Liquid	21.2	27.9	50.9
		Pseudowollastonite	<0.34	49.9	49.7
1300	2	"$CuO_{0.5}$"-poor liquid	39.3	23.3	37.4
		*"$CuO_{0.5}$"-rich liquid	94.4	1.2	4.4
		Pseudowollastonite	<0.77	49.6	49.6
1300	2	"$CuO_{0.5}$"-poor liquid	39.8	22.8	37.3
		*"$CuO_{0.5}$"-rich liquid	95.6	0.00	4.4
		Pseudowollastonite	<0.17	49.8	50.0
1250	1	"$CuO_{0.5}$"-poor liquid	49.3	16.6	34.1
		*"$CuO_{0.5}$"-rich liquid	91.1	1.2	7.7
		Pseudowollastonite	<0.77	49.6	49.6
1250	2	Liquid	27.2	22.7	50.1
		Pseudowollastonite	<0.34	49.8	49.8
1225	2	"$CuO_{0.5}$"-poor liquid	57.0	12.3	30.7
		*"$CuO_{0.5}$"-rich liquid	86.7	2.4	10.9
		Pseudowollastonite	<0.26	49.9	49.8
1200	1	Liquid	54.8	12.3	32.9
		Cuprite	99.9	0.00	0.1

*according to Liu [16] - semi-quantitative estimates with uncertainties that were not established.

silicate (Ca_2SiO_4) (4h), and the coexistence of two liquids (4i).

In several microstructures (Fig. 4a, c, e, f) the solid phases presented are surrounded by an "outer layer" of crystals. The analysis of these layers showed that their compositions lie on the tie line between the cristobalite (SiO_2) and bulk slag (Fig. 4a) or cuprite (Cu_2O) and bulk slag (Fig. 4f); this indicated that the appearance of these metastable phases was the result of the quenching process and did not affect the whole equilibria within the samples. The advantage of the rapid cooling in characterizing these systems is illustrated in Fig. 4g, which shows regions of slow cooling, characterised by partially crystallised (microcrystalline) slag, with the rapidly cooled slag, exhibiting amorphous material homogeneous in chemical composition.

The identities of the tridymite and cristobalite (SiO_2) phases were distinguished based on their different crystal morphologies. Fig. 4a shows the rounded liquid/crystal interface in the case of the cubic high temperature form of cristobalite in contrast to the faceted plate-shaped tridymite (SiO_2) crystals formed at low temperature (Fig. 4b, e, j). The change in the shape of the crystals was found to occur in the temperature interval from 1400 to 1500 °C, which agreed with the tridymite (SiO_2) – cristobalite (SiO_2) transition temperature reported by Jak et al. [47] as 1465 °C. The correspondence between the observed shape of the crystal and the phase (tridymite or cristobalite (SiO_2) was confirmed using X-ray powder diffraction of two samples of the same composition equilibrated at 1400 and 1500 °C. Fig. 5 shows the comparison of the XRD patterns of the samples and their microstructures. The composition of phases present in these samples are given in Table 1 and plotted in Fig. 6. The compositions of liquid phase were not used for the construction of the corresponding isotherms but were given for the presentation purposes. Cristobalite (SiO_2) identified together with tridymite (SiO_2) in the sample equilibrated at lower temperature is considered metastable remaining due to very small driving force (Gibbs energy difference) of transition and still allows to correlate visual appearance of crystals with the corresponding phases for the purposes of this study.

The mean compositions of the phases obtained in the present study and the oxygen partial pressures corresponding to these compositions, calculated using Factsage 7.3 FactPS [25] and internal thermodynamic

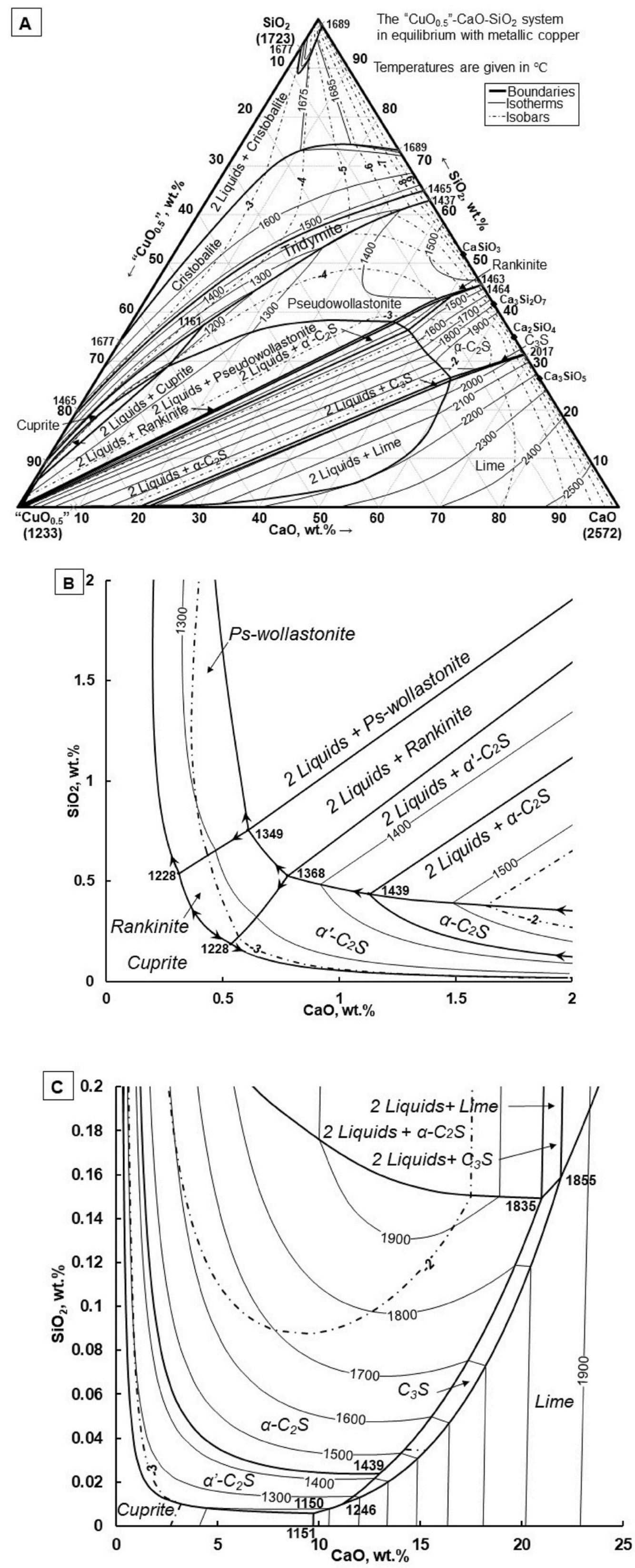

Fig. 7. Liquidus surface of the "$CuO_{0.5}$"-CaO-SiO_2 system in equilibrium with metallic copper: (a) Full liquidus surface; (b) detailed "$CuO_{0.5}$" corner of the diagram; and (c) detailed area close to the "$CuO_{0.5}$"-CaO binary. Temperatures are given in °C. [2.11 2.13].

database [8], are reported in Table 2. Table 3 presents the data on this system reported by Liu [16]. The composition of metallic copper in every sample (except those equilibrated at Ir wire) was found to be > 99.9 mol.% Cu; for the presentation purposes the results of those measurements were not included in Table 2. The word "liquid" in Tables 2 and 3 then stands for the oxide liquid. The concentrations of impurities (FeO_x and $AlO_{1.5}$) in the liquid slag were measured for every sample and their sum did not exceed 0.06 mol.% on average; for the presentation purposes the results of those measurements were not included in Table 2. These data, in conjunction with the established data for the binary systems [18,23] and the experimental results reported by Hidayat et al. [15], are used for the construction of the "$CuO_{0.5}$"--CaO-SiO_2 system liquidus surface shown in Fig. 6. Phase relations and isotherms in the CaO-rich region, where no experimental data are available, are those predicted by Hidayat and Jak [24], and are shown in the dashed lines.

The liquidus in the cristobalite and tridymite (SiO_2) primary phase fields was measured in the present study at temperatures ranging from 1160 °C to 1706 °C. The temperature of the tridymite-cristobalite (SiO_2) transition was not determined directly and was accepted from Hidayat and Jak [24] as 1465 °C. The SiO_2-rich miscibility gap constructed from the new experimental data follows the trend predicted by Hidayat [24] but there are differences in the compositions of the actual and previously predicted boundaries. The cristobalite (SiO_2) primary phase field is bounded by the monotectic reaction, that is, with the formation of the liquid-liquid immiscibility. The maximum concentration of "$CuO_{0.5}$" measured in the SiO_2-rich liquid was 7.9 mol.% (9.3 wt%), at a temperature of 1675 °C compared to approximately 3 mol.% (3.6 wt%) "$CuO_{0.5}$" previously suggested by the model predictions [24]. The composition of the SiO_2-poor liquid ranged from 37.8 mol.% (33.8 wt% SiO_2) at 1677 °C on "$CuO_{0.5}$" - SiO_2 binary to 71.8 mol.% (73.2 wt%) at 1689 °C on the CaO–SiO_2 binary. The experimental data obtained for the liquidus isotherms at 1650, 1600 and 1500 °C in the cristobalite (SiO_2), and 1400 and 1300 °C in the tridymite (SiO_2) primary phase fields, are consistent with those previously reported at 1200 and 1300 °C by Hidayat et al. [10]. The boundary line between the tridymite (SiO_2) and pseudowollastonite ($CaSiO_3$) primary phase fields was corrected based on the new data, expanding the tridymite (SiO_2) primary phase field.

At 1160 °C, the system was found to be completely solid (see Fig. 4j); the closest temperature at which liquid was found among the phases was 1162 °C (see Fig. 4f). Thus, the temperature of the liquid ↔ pseudowollastonite ($CaSiO_3$) + cuprite (Cu_2O) + tridymite (SiO_2) eutectic, based on the current data, is between 1160 and 1162 °C. The composition of this eutectic liquid based on the intersection of the pseudowollastonite ($CaSiO_3$) + tridymite (SiO_2), pseudowollastonite ($CaSiO_3$) + cuprite (Cu_2O) and cuprite (Cu_2O) + tridymite (SiO_2) boundary lines was determined to be 45.6 mol.% "$CuO_{0.5}$", 14.1 mol.% CaO and 40.3 mol.% SiO_2 (50.4 wt% "$CuO_{0.5}$", 12.2 wt% CaO and 37.4 wt% SiO_2).

The liquid-liquid immiscibility in the low-silica region was found to cover the cuprite (Cu_2O), pseudowollastonite ($CaSiO_3$), rankinite ($Ca_3Si_2O_7$), dicalcium silicate (Ca_2SiO_4) and predicted to reach tricalcium silicate (Ca_3SiO_5) and lime (CaO) primary phase fields. The boundaries for 2 liquids + cuprite (Cu_2O), and 2 liquids + pseudowollastonite ($CaSiO_3$) were taken from the data by Liu [11]. Since the high temperatures associated the limits of the liquid immiscibilities in the region having CaO/SiO_2 > 60/40 mol.% could not be achieved with the current equipment these values were taken to be those predicted by Hidayat and Jak [24] and corrected based on the new data on the phase equilibria between two immiscible liquids and dicalcium silicate (Ca_2SiO_4). The immiscibility in the high CaO region is consistent with the findings for the "$CuO_{0.5}$" -CaO-SiO_2 system at 7 wt% FeO reported by Takeda [48].

The liquidus of rankinite ($Ca_3Si_2O_7$) was determined on the basis of the results obtained for the pseudowollastonite ($CaSiO_3$) + rankinite ($Ca_3Si_2O_7$) boundary line and dicalcium silicate (Ca_2SiO_4) liquidus. It was found that the rankinite ($Ca_3Si_2O_7$) primary phase field is characterized by a narrow interval of stability.

The "$CuO_{0.5}$" concentrations in the cristobalite and tridymite (SiO_2) measured using L_α lines after further custom corrections (equations (4) and (1–3)) were found to be close to zero and below the EPMA detection limit. The "$CuO_{0.5}$" concentrations in the pseudowollastonite ($CaSiO_3$), rankinite ($Ca_3Si_2O_7$) and dicalcium silicate (Ca_2SiO_4) measured using L_α lines with further custom corrections (equations (4) and (1–3)) were found to be in the range of 0.07–0.14 mol.% (0.09–0.17 wt%), 0.1–0.13 mol.% (0.12–0.16 wt%) and 0.26–0.9 mol.% (0.32–1.1 wt%) "$CuO_{0.5}$" respectively. These values are lower than the 0.7–1.1 mol.% (0.83–1.3 wt%) "$CuO_{0.5}$" in tridymite (SiO_2) and 0.2–0.9 mol.% (0.21–1.1 wt%) "$CuO_{0.5}$" in pseudowollastonite ($CaSiO_3$) reported previously [15,16] based on the EPMA measurements using Cu-K_α line (8.05 keV) that appear to overestimate values due to the secondary X-ray fluorescence effects.

The mixture of $Ca_4Si_6O_{16}$ and SiO_2 was equilibrated together with metallic copper powder at 1450 °C and $p(O_2) = 10^{-11.5}$ during this study. The concentration of copper was found to be negligible (0.07 ± 0.03 mol.% "$CuO_{0.5}$" with the detection limit of 0.04 mol.%). This total copper is a sum of all possible species, i.e. Cu^0, $CuO_{0.5}$ and CuO. Out of these species, Cu^0 is expected to be independent on $p(O_2)$, $CuO_{0.5}$ (the major species) to have a slope $\log(CuO_{0.5}) = 0.25\log p(O_2) + \text{const}$, and CuO to have a slope $\log(CuO) = 0.5\log p(O_2) + \text{const}$ – the most suppressed by low $p(O_2)$ and therefore negligible. This confirms that Cu^0 in slag does not exceed 0.07 ± 0.03 mol.%, is suppressed by CaO and SiO_2 at high concentrations of these components.

Using the present and previous experimental data on the binary [18, 23] and ternary [15,16] systems, the projected liquidus surface of the "$CuO_{0.5}$"-CaO-SiO_2 system in equilibrium with metallic copper was constructed and shown in Fig. 7; the isobars presented are those calculated using Factsage 7.3 FactPS [25] and internal thermodynamic database [8].

4. Conclusions

The study has produced new experimental data on phase equilibria in the "$CuO_{0.5}$"-CaO-SiO_2 system in equilibrium with metallic copper in the temperature range from 1160 °C to 1706 °C, including: the primary phase fields of tridymite and cristobalite (SiO_2) and pseudowollastonite ($CaSiO_3$), boundary lines between them and between pseudowollastonite ($CaSiO_3$), rankinite ($Ca_3Si_2O_7$) and dicalcium silicate (Ca_2SiO_4) primary phase fields, as well as the two liquid miscibility gap in the high-silica area of the diagram.

The system was shown to contain an invariant eutectic liquid ↔ cuprite (Cu_2O) + pseudowollastonite ($CaSiO_3$) + tridymite (SiO_2) at 45.6 mol.% "$CuO_{0.5}$", 14.1 mol.% CaO and 40.3 mol.% SiO_2 (50.4 wt% "$CuO_{0.5}$", 12.2 wt% CaO and 37.4 wt% SiO_2) at approximately 1161 ± 3 °C. The two liquid miscibility gap in the low-silica region of the diagram is predicted to extend from the copper oxide rich melts, containing almost 99 mol.% "$CuO_{0.5}$", to copper oxide concentrations of ~9 mol.% (~11 wt%) "$CuO_{0.5}$" for CaO/SiO_2 ratios between 60/40 and 70/30 mol.%.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to thank Australian Research Council, linkage project LP190101020 "Future copper metallurgy for the age of e-mobility and the circular economy", and industry partners: Anglo American Pt (South Africa), Aurubis AG (Germany), BHP Olympic Dam (Australia), Boliden (Sweden), Glencore Technology (Australia), Outotec Oy (Finland), Peñoles (Mexico), RHI Magnesita (Austria), Rio Tinto Kennecott (USA), Umicore NV (Belgium) for providing funding and technical support.

References

[1] B. Gorai, R.K. Jana, Premchand, Characteristics and utilization of copper slag - a review, Resour. Conserv. Recycl. 39 (2003) 299–313, https://doi.org/10.1016/S0921-3449(02)00171-4.

[2] M.E. Schlesinger, M.J. King, K.C. Sole, W.G.I. Davenport (Eds.), Extractive Metallurgy of Copper, Elsevier, Amsterdam, Neth., 2011.

[3] A. Malfliet, S. Lotfian, L. Scheunis, V. Petkov, L. Pandelaers, P.T. Jones, et al., Degradation mechanisms and use of refractory linings in copper production processes: a critical review, J. Eur. Ceram. Soc. 34 (2014) 849–876, https://doi.org/10.1016/j.jeurceramsoc.2013.10.005.

[4] A. Fallah Mehrjardi, P.C. Hayes, T. Azekenov, L. Ushkov, E. Jak, Phase chemistry study of the interactions between slag and refractory in coppermaking processes, in: The 10th International Conference on Molten Slags, Fluxes and Salts (MOLTEN16), Seattle, Washington, USA, 2016, pp. 1071–1076, https://doi.org/10.1002/9781119333197.ch114.

[5] A. Fallah Mehrjardi, P.C. Hayes, T. Azekenov, L. Ushkov, E. Jak, Investigation of chemical interactions between slag and refractories in copper production processes, in: The 9th International Copper Conference (Copper2016), 2016. Kobe, Japan.

[6] A. Fallah Mehrjardi, P.C. Hayes, T. Azekenov, L. Ushkov, E. Jak, A systematic approach to the development of self-healing slag refractory systems, in: The Unified International Technical Conference on Refractories, 2017. Santiago, Chile.

[7] A. Fallah-Mehrjardi, P.C. Hayes, E. Jak, The effect of CaO on gas/slag/Matte/tridymite equilibria in fayalite-based copper smelting slags at 1473 K (1200 °C) and P(SO_2) = 0.25 atm, Metall. Mater. Trans. B 49 (2018) 602–609, https://doi.org/10.1007/s11663-018-1170-6.

[8] M. Shevchenko, D. Shishin, P.C. Hayes, E. Jak, Recent developments in experimental and thermodynamic modelling techniques for the characterisation of phase equilibria in complex high temperature oxide systems, in: The 11th International Conference on Molten Slags, Fluxes and Salts, 2021. Virtual, Korea.

[9] D. Shishin, S.A. Decterov, Critical assessment and thermodynamic modeling of Cu-O and Cu-O-S systems, Calphad 38 (2012) 59–70.

[10] A. Yazawa, Extractive metallurgical chemistry with special reference to copper smelting, in: 28th Congress of IUPAC Vancouver, 1981, pp. 1–21.

[11] T. Hidayat, D. Shishin, S. Decterov, E. Jak, Critical assessment and thermodynamic modeling of the Cu-Fe-O-Si system, Calphad 58 (2017) 101–114.

[12] D. Shishin, T. Hidayat, E. Jak, Thermodynamic assessment of the CaO–Cu_2O–FeO–Fe_2O_3 system, Calphad 68 (101715) (2020), https://doi.org/10.1016/j.calphad.2019.101715.

[13] U. Kuxmann, K. Kurre, Miscibility gap in the system copper-oxygen and its alteration by the oxides CaO, SiO_2, Al_2O_3, MgO·Al_2O_3, and ZrO_2, Erzmetall 21 (1968) 199–209.

[14] C. Landolt, Equilibrium Studies in the System Copper–silicon–oxygen, Ph.D. Thesis, Pennsylvania State Univ, 1969.

[15] T. Hidayat, H.M. Henao, P.C. Hayes, E. Jak, Phase equilibria studies of Cu-O-Si systems in equilibrium with air and with metallic copper, and Cu-Me-O-Si systems (Me = Ca, Mg, Al, and Fe) in equilibrium with metallic copper, Metall. Mater. Trans. B 43 (2012) 1290–1299.

[16] X.G. Liu, Experimental Phase Equilibria Studies in Oxide Systems for Copper Smelting Slags, Master of Philosophy, University of Queensland, Australia, 2012.

[17] L. Xia, Z. Liu, P.A. Taskinen, Equilibrium study of the Cu-O-SiO_2 system at various oxygen partial pressures, J. Chem. Thermodyn. 98 (2016) 126–134, https://doi.org/10.1016/j.jct.2016.03.023.

[18] M. Shevchenko, E. Jak, Experimental liquidus studies of the binary Pb-Cu-O and ternary Pb-Cu-Si-O systems in equilibrium with metallic Pb-Cu alloys, J. Phase Equilibria Diffus. 40 (2019) 671–685, https://doi.org/10.1007/s11669-019-00754-8.

[19] Y. Takeda, Phase diagram of CaO-FeO-Cu_2O slag under copper saturation, in: Metall. Mater. Process.: Princ. Technol., Yazawa Int. Symp. Minerals vol. 1, Metals & Materials Society, Warrendale, PA, 2003, pp. 211–225.

[20] P. Coursol, A.D. Pelton, M. Zamalloa, Phase equilibria and thermodynamic properties of the Cu_2O-CaO-Na_2O system in equilibrium with copper, Metall. Mater. Trans. B 34B (2003) 631–638.

[21] J. Hamuyuni, N. Hellsten, G. Akdogan, P. Taskinen, The liquidus in Cu-O-CaO system at metallic copper saturation up to 1698 K, J. Am. Ceram. Soc. 98 (2015) 320–323, https://doi.org/10.1111/jace.13282.

[22] G. Eriksson, P. Wu, M. Blander, A.D. Pelton, Critical evaluation and optimisation of the thermodynamic properties and phase diagrams of the MnO-SiO_2 and CaO-SiO_2 systems, Can. Metall. Q. 33 (1994) 13–21.

[23] S. Cheng, M. Shevchenko, P.C. Hayes, E. Jak, Experimental phase equilibria studies in the FeO-Fe_2O_3-CaO-SiO_2 system and the subsystems CaO-SiO_2, FeO-Fe_2O_3-SiO_2 in air, Metall. Mater. Trans. B (2021), https://doi.org/10.1007/s11663-021-02159-w. Ahead of Print.

[24] T. Hidayat, E. Jak, Thermodynamic modeling of the "Cu_2O"-SiO_2, "Cu_2O"-CaO, and "Cu_2O"-CaO-SiO_2 systems in equilibrium with metallic copper, Int. J. Mater. Res. 105 (2014) 249–257.

[25] C.W. Bale, E. Belisle, P. Chartrand, S.A. Decterov, G. Eriksson, A.E. Gheribi, et al., FactSage thermochemical software and databases, 2010-2016, Calphad 54 (2016) 35–53, https://doi.org/10.1016/j.calphad.2016.05.002.

[26] T. Hidayat, H.M. Henao, P.C. Hayes, E. Jak, Phase equilibria studies of the Cu-Fe-O-Si system in equilibrium with air and with metallic copper, Metall. Mater. Trans. B 43 (2012) 1034–1045.

[27] E. Jak, P.C. Hayes, H.-G. Lee, Improved methodologies for the determination of high temperature phase equilibria, Korean J. Mineral. Mater. Inst. (Seoul) 1 (1995) 1–8.

[28] E. Jak, Integrated experimental and thermodynamic modelling research methodology for metallurgical slags with examples in the copper production field, in: 9th Intl. Conf. On Molten Slags, Fluxes and Salts (MOLTEN12), The Chinese Society for Metals, Beijing, China, 2012.

[29] T. Abe, B. Sundman, H. Onodera, Thermodynamic assessment of the Cu-Pt system, J. Phase Equilibria Diffus. 27 (2006) 5–13, https://doi.org/10.1361/105497196x92736.

[30] D.J. Chakrabarti, D.E. Laughlin, The Cu-Ir (Copper-Irridium) system, Bull. Alloy Phase Diag. 8 (1987) 132–136, https://doi.org/10.1007/BF02873198.

[31] L.H. Pizetta Zordão, V.A. Oliveira, G.E. Totten, L.C.F. Canale, Quenching power of aqueous salt solution, Int. J. Heat Mass Tran. 140 (2019) 807–818, https://doi.org/10.1016/j.ijheatmasstransfer.2019.06.036.

[32] W. Luty, Types of cooling Media and their properties, in: B. Liščić, H.M. Tensi, W. Luty (Eds.), Theory and Technology of Quenching: A Handbook, Springer Berlin Heidelberg, Berlin, Heidelberg, 1992, pp. 248–340.

[33] J. Philibert, Method for calculating the absorption correction in electron-probe microanalysis, 3rd Intl. Symp., in: X-Ray Opt. X-Ray Microanal, Stanford Univ, 1963, pp. 379–392.

[34] P. Duncumb, S.J.B. Reed, Calculation of stopping power and backscatter effects in electron probe microanalysis, 1968, pp. 133–154.

[35] P. Duncumb, Quantitative electron probe microanalysis, Electron Microsc. Anal., Proc. Anniv. Meet. 25th (1971) 132–137.

[36] A. Fallah-Mehrjardi, T. Hidayat, P.C. Hayes, E. Jak, Experimental investigation of gas/slag/Matte/tridymite equilibria in the Cu-Fe-O-S-Si system in controlled gas atmospheres: development of technique, Metall. Mater. Trans. B 48 (2017) 3002–3016, https://doi.org/10.1007/s11663-017-1073-y.

[37] M. Shevchenko, E. Jak, Experimental liquidus studies of the Pb-Fe-Si-O system in equilibrium with metallic Pb, Metall. Mater. Trans. B 49 (2018) 159–180, https://doi.org/10.1007/s11663-017-1136-0.

[38] M. Shevchenko, E. Jak, Experimental phase equilibria studies of the PbO-SiO_2 system, J. Am. Ceram. Soc. 101 (2017) 458–471, https://doi.org/10.1111/jace.15208.

[39] M. Shevchenko, J. Chen, E. Jak, Establishing additional correction for quantitative EPMA measurements in the system PbO-SiO_2, Brisbane, Australia, in: AMAS 2017, 14th Biennial Australian Microbeam Analysis Symposium, QUT, Brisbane, 2017, pp. 94–95.

[40] T. Kawasaki, H. Ishizuka, Experimental study of Fe^{3+} solubility in cristobalite and its application to a metamorphosed quartz-magnetite rock from Mt. Riiser-Larsen area, Napier Complex, East Antarctica, J. Mineral. Petrol. Sci. 103 (2008) 255–265, https://doi.org/10.2465/jmps.080216.

[41] X. Llovet, P.T. Pinard, J.J. Donovan, F. Salvat, Secondary fluorescence in electron probe microanalysis of material couples, J. Phys. Appl. Phys. 45 (2012), https://doi.org/10.1088/0022-3727/45/22/225301, 225301/225301-225301/225312.

[42] M. Shevchenko, E. Jak, Experimental liquidus studies of the Pb-Fe-Si-O system in air, J. Phase Equilibria Diffus. 40 (2019) 319–355, https://doi.org/10.1007/s11669-019-00727-x.

[43] M. Shevchenko, E. Jak, Experimental liquidus studies of the Zn-Fe-Si-O system in air, IJMR (Int. J. Mater. Res.) 110 (2019) 600–607, https://doi.org/10.3139/146.111779.

[44] M. Shevchenko, E. Jak, Experimental Liquidus Studies of the ternary Zn-Cu-Si-O System in Equilibrium with Metallic Zn-Cu Alloys, Private communication, 2018.

[45] M. Shevchenko, E. Jak, Experimental liquidus study of the binary PbO-ZnO and ternary PbO-ZnO-SiO_2 systems, Ceram. Int. 45 (2019) 6795–6803, https://doi.org/10.1016/j.ceramint.2018.12.172.

[46] M. Shevchenko, T. Hidayat, P. Hayes, E. Jak, Liquidus of “FeO”-SiO_2-PbO slags in equilibrium with air and with metallic lead, in: Molten 2016, 10th Intl. Conf. On Molten Slags, Fluxes and Salts, 2016, pp. 1221–1228, https://doi.org/10.1002/9781119333197.ch131. Seattle, Washington, USA.

[47] E. Jak, S.A. Decterov, P. Wu, P.C. Hayes, A.D. Pelton, Thermodynamic optimisation of the systems PbO-SiO_2, PbO-ZnO, ZnO-SiO_2 and PbO-ZnO-SiO_2, Metall. Mater. Trans. B 28B (1997) 1011–1018.

[48] Y. Takeda, Miscibility gap in the calcia-silica-copper oxide (Cu_2O)-iron oxide (Fe_3O_4) system under copper saturation and distribution of impurities, Mater. Trans., JIM 34 (1993) 937–945.